

Core Chip: ESP32-C3-MINI-1-N4 | Protocol Support: UART-WiFi-001, SPI-WiFi-002, CAN-WiFi-003, RS485-WiFi-004 to WiFi.

1. Design Architecture Reference

This manual covers four dedicated transparent transmission modules based on the **ESP32-C3-MINI-1-N4** core. These modules are designed to seamlessly integrate industrial fieldbus data (RS485/CAN/SPI/UART) into wireless networks or cloud platforms.

- **Core Processor:** ESP32-C3-MINI-1-N4 (RISC-V @160MHz), featuring integrated 4MB Flash and 2.4GHz WiFi.
- **Platform Design:** A unified wireless engine with dedicated signal processing circuits tailored for different buses.
- **Physical Layer Adaptation:**
 - **CAN Module:** Integrates the **TCAN3414DR**, supporting the ISO 11898-2 standard.
 - **RS485 Module:** Integrates a 3.3V differential transceiver with hardware-level automatic direction switching.
 - **SPI/UART Module:** Compatible with 3.3V and 5V data levels; no level conversion is required.
- **Communication modes:** Supports **AP Mode** (for device configuration) and **STA Mode** (for network transmission).

2. Module Specifications & Interface Definitions

Module Model	Physical Layer Chip	Key Pin Assignment (ESP32-C3)	Interface Features
C3-CAN	TCAN3414DR	TX: GPIO2 / RX: GPIO3	Supports CAN 2.0B, max 1Mbps, strong ESD protection
C3-485	SP3485EN-L_TR	TX: GPIO4 / RX: GPIO5 / DE: GPIO6	Supports Modbus RTU transparent transmission

Module Model	Physical Layer Chip	Key Pin Assignment (ESP32-C3)	Interface Features
C3-UART	TTL Header	TX: GPIO4 / RX: GPIO5	Default: 115200, 8, N, 1
C3-SPI	TTL Header	SCK/MOSI/MISO/CS/REQ: GPIO 4,5,6,7,10	Module operates in Slave Mode

3. LED Indicator Status Description

Monitor the board's LED to determine its current operational state:

- **1s Slow Flash (1s):** The device is in **AP Configuration Mode**.
- **200ms Fast Flash: WiFi connection** successful.
- **500ms Medium Flash:** Server connection successful; device is ready for transparent transmission.

4. Quick Configuration Guide

Option A: Web Page Configuration (Recommended)

1. **Power & Status:** Connect the power. If the LED flashes slowly every 1 second, it is in AP mode.
 - **Force Entry:** Long-press the **Config** button for over 3 seconds until the LED flashes at 1-second intervals.
2. **Connect to Hotspot:** Use a phone or PC to connect to the hotspot named SmartUART_Config (or the corresponding protocol name). The password is 12345678.
3. **Access Backend:** Open a browser and navigate to **192.168.8.1**.
4. **Set Parameters:**
 - **SSID:** Your WiFi Name.
 - **Password:** Your WiFi Password.
 - **Server IP:** The IP address of your server (e.g., 192.168.2.18).
 - **Port:** Target listening port of the server (e.g., 8888).

- **Baudrate:** Baud rate (e.g., 115200) and data bits (data bits option only appears for UART/RS485).
- 5. **Save & Apply:** Click **Save & Connect** to send the configuration.
- 6. **Status Indication:** An LED 200ms fast flash indicates WiFi is connected; a 500ms medium flash indicates the server is connected.

Option B: USB Setup Tool (For Batch Configuration)

1. **Download Tool:** Visit <https://github.com/wnettech/WNetTech-Tool>.
2. **Select Software:**
 - **Chinese Version:** WNetTool_Config_v1.0.0_cn.exe.
 - **English Version:** WNetTool_Config_v1.0.0_en.exe.
3. **Environment Preparation:** It is recommended to run the tool from the folder C:\My_Project_Tools.
4. **Set Parameters:** Enter the SSID, Password, Server IP, Port, and Baudrate as described in Option A.

5. Transparent Data Transmission Test

1. **Prepare Server:** Download the server test software from <https://github.com/wnettech/WNetTech-Tool>. Use WNetTool_Server_Port55001_v1.0.0_en.exe (or _cn.exe).
2. **Get Local IP:** Enter `ipconfig` in the command prompt to find your IPv4 address (e.g., 192.168.18.4).
3. **Configure Settings:**
 - Set the device **Server IP** to your PC's IP (e.g., 192.168.18.4).
 - Set the **Port** to 55001 (required by the test software).
 - Ensure Wi-Fi credentials are correct
 - Ensure Baudrate(If Model is Uart or RS485) is correct.

After successful setup, the device will enter a 200ms flash (if the Wi-Fi settings are correct).

4. **Running the Test:**
 - Once configured correctly, the device LED will flash at 200ms.

- Launch the [WNetTool_Server_Port55001_v1.0.0](#) software.
 - The LED will transition from a 200ms fast flash to a **500ms medium flash**, indicating a successful server connection.
5. **Data Verification:** Using the UART module as an example, open a serial assistant with matching baud rate settings. Send [Hello WiFi](#); the PC test software should immediately display the original data in its message receiving window.